IGCSE 0607 Practice Questions — E3.2

Sketching graphs on a calculator

Total Questions: 18 — **Total Marks:** 135 Marks

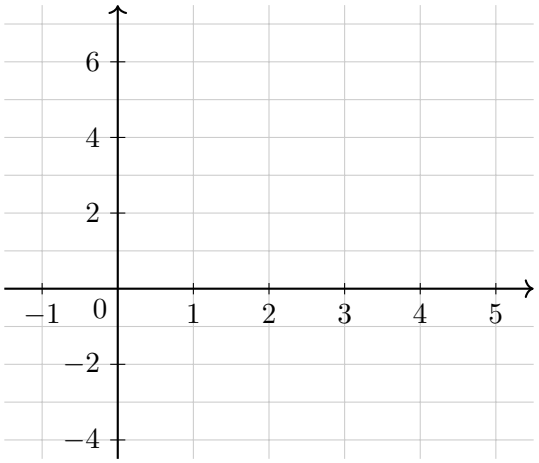
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- **E3.2** Sketching graphs on a calculator
  → **Paper 4 (Calculator)** *(18 Qs)*

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1 Consider the function $f(x) = x^2 - 4x + 1$ for $-1 \leq x \leq 5$.

(a) Use your graphic display calculator to sketch the graph of $y = f(x)$ on the axes provided.



[2]

(b) Write down the coordinates of the vertex of the graph.

(..... , ...) **[1]**

(c) Find the zeros of $f(x)$.
Give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[2]**

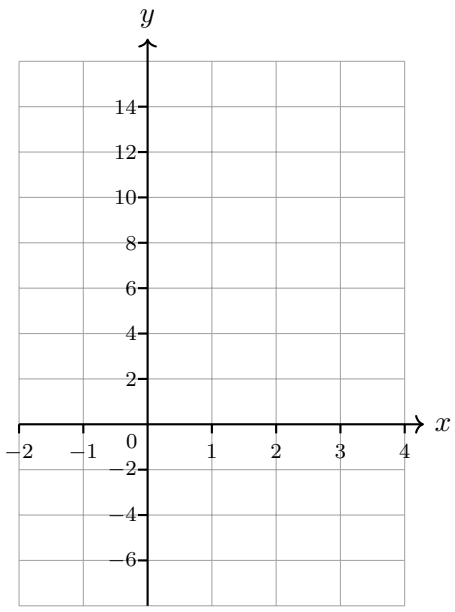
2 Consider the function $f(x) = 2x^2 - 5x - 3$ for $-2 \leq x \leq 4$.

(a) Use your graphic display calculator to complete the table of values for $f(x)$.

x	-2	-1	0	1	2	3	4
$f(x)$	15		-3		-5		9

[2]

(b) On the grid, plot the points from your table and draw a smooth curve to show the graph of $y = f(x)$ for $-2 \leq x \leq 4$.



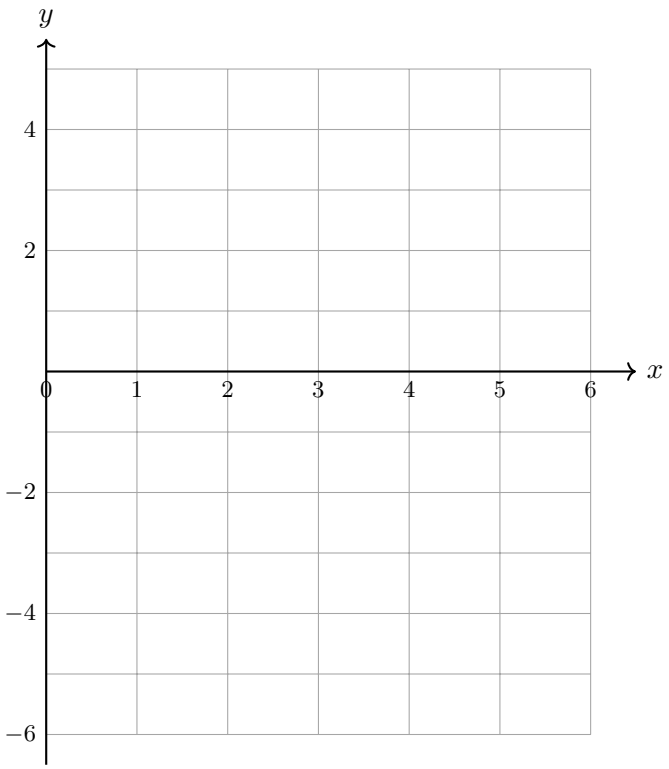
[2]

(c) Use your graphic display calculator to find the coordinates of the vertex of the graph of $y = f(x)$.

(..... , ...) **[1]**

3 Let $f(x) = -x^2 + 6x - 5$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $0 \leq x \leq 6$.



[2]

(b) Find the zeros of $f(x)$.

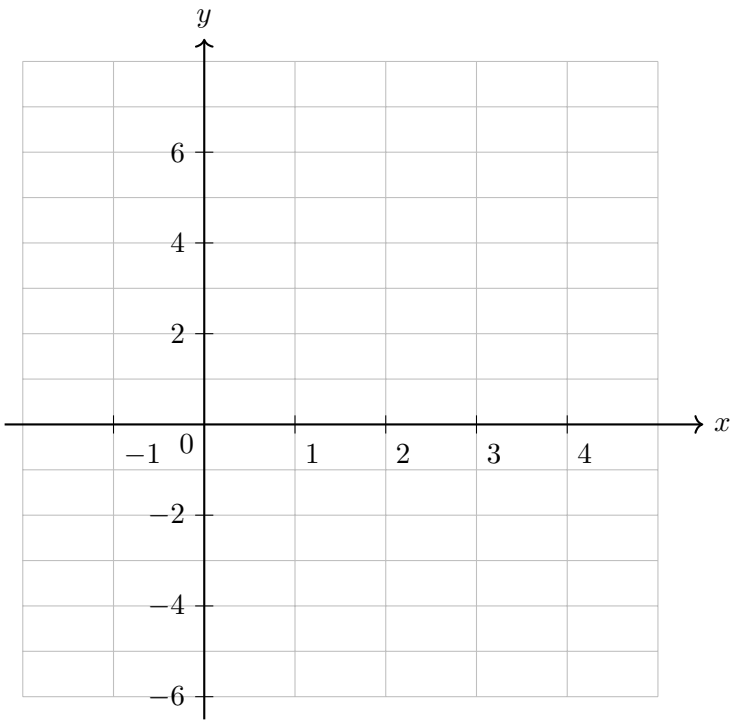
$x =$ ..... , $x =$ ..... **[2]**

(c) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

4 Let $f(x) = -2x^2 + 5x + 3$.

(a) On the axes, sketch the graph of $y = f(x)$ for $-1.5 \leq x \leq 4$.



[2]

(b) Find the coordinates of the local maximum.

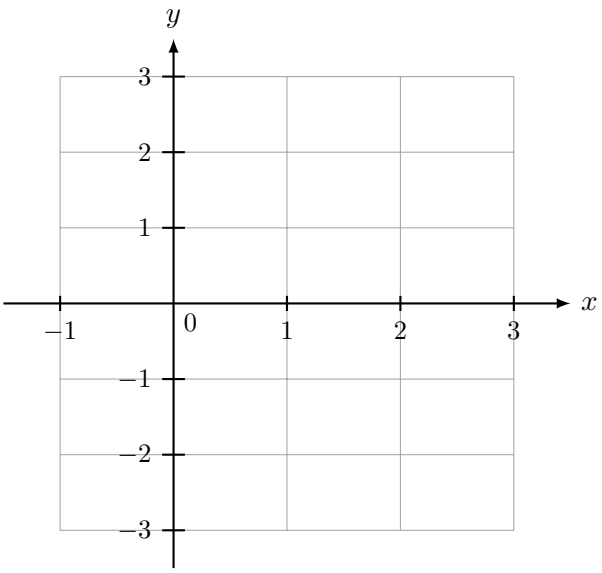
(..... , ...) **[2]**

(c) Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

5 Consider the function $f(x) = x^3 - 3x^2 + 2$ for $-1 \leq x \leq 3$.

(a) On the grid, sketch the graph of $y = f(x)$.



[2]

(b) Find the zeros of $f(x)$.

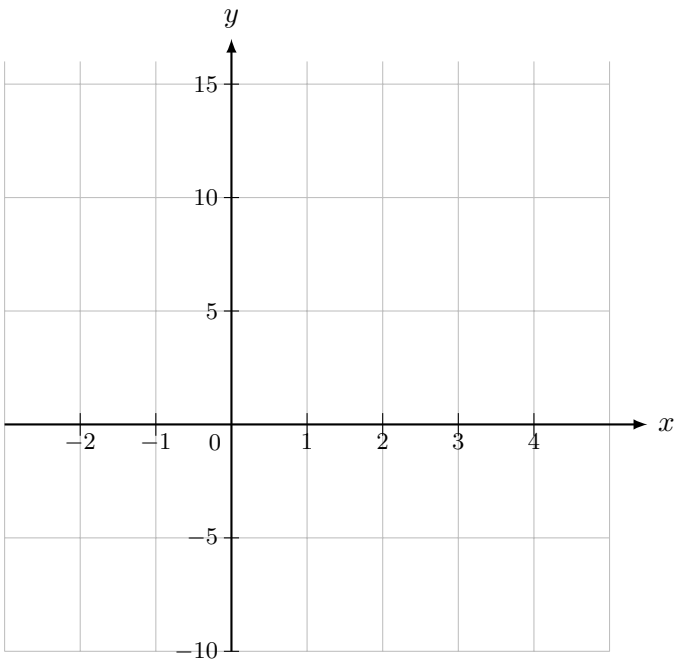
$x =$ ..... **[2]**

(c) Find the coordinates of the local minimum point of the graph.

(..... , ...) **[2]**

6 A function is given by $f(x) = 2x^2 - 5x - 3$ for the domain $-2 \leq x \leq 4$.

(a) On the axes, sketch the graph of $y = f(x)$.



[2]

(b) Write down the zeros of $f(x)$.

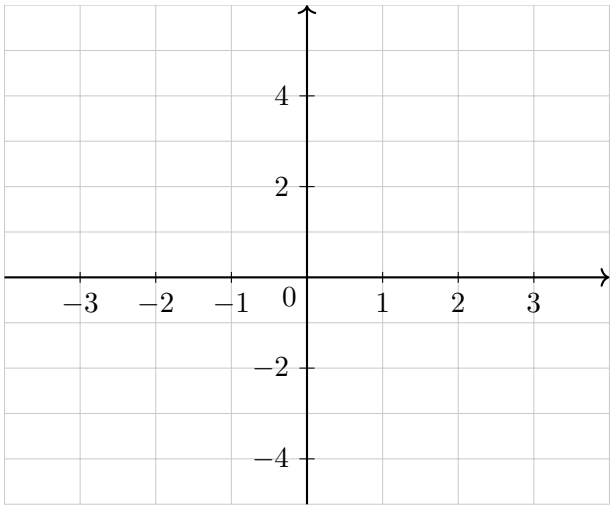
$x =$ ..... , $x =$ ..... **[2]**

(c) Write down the coordinates of the local minimum.

(..... , ...) **[2]**

7 Consider the function $f(x) = \dfrac{x^3}{4} - 2x + 1$ for $-3.5 \leq x \leq 3.5$.

(a) On the grid, sketch the graph of $y = f(x)$.



[2]

(b) Find the coordinates of the local maximum point and the local minimum point.
Give your answers correct to 2 decimal places.

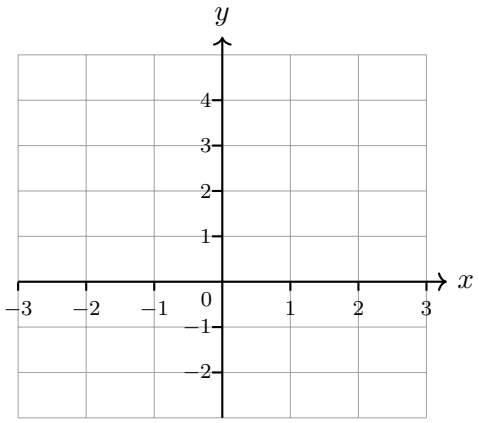
Local maximum = ... , Local minimum = ... **[2]**

(c) Write down the range of values of k for which the equation $f(x) = k$ has three distinct real solutions.

..... **[2]**

8 Consider the functions $f(x) = \dfrac{4}{x^2+1}$ and $g(x) = 2x - 1$ for $-3 \leq x \leq 3$.

(a) Use your graphic display calculator to sketch the graphs of $y = f(x)$ and $y = g(x)$ on the grid.



[3]

(b) Find the coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = g(x)$. Give your answers correct to 3 significant figures.

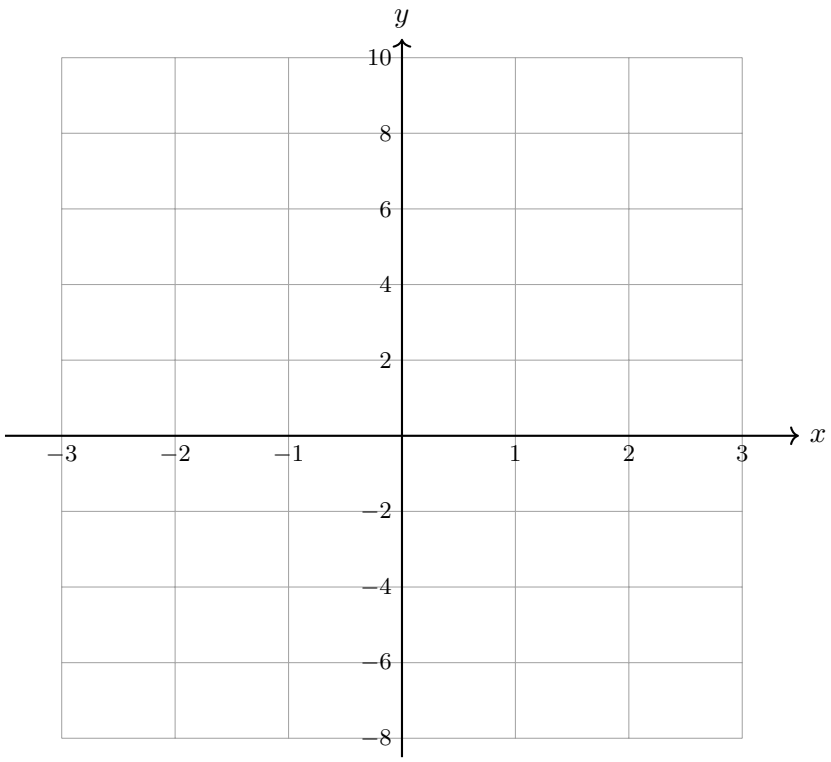
(..... , ...) **[2]**

(c) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

9 Let $f(x) = x^3 - 3x + 1$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.



[2]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

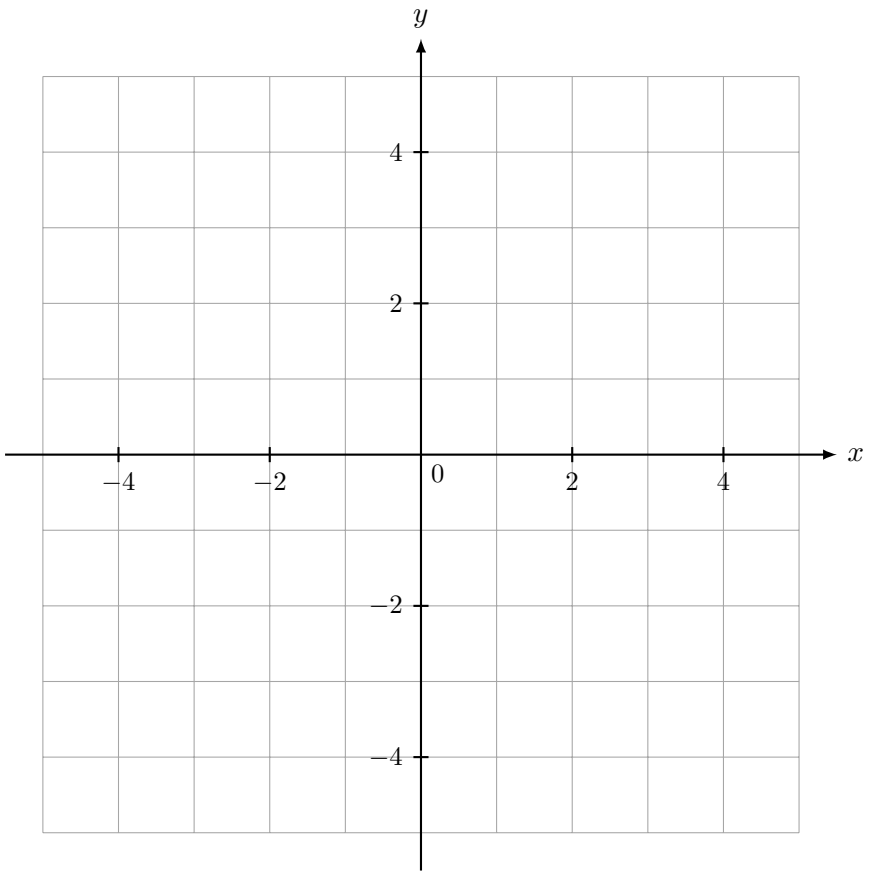
(c) On the same axes, sketch the graph of $y = 2x - 1$. **[1]**

(d) Find the x-coordinates of the points of intersection of the two graphs.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[2]**

10 Let $f(x) = \dfrac{2}{x+1}$ and $g(x) = x - 2$ for $-5 \leq x \leq 5$.

(a) On the grid, sketch the graphs of $y = f(x)$ and $y = g(x)$.



[3]

(b) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

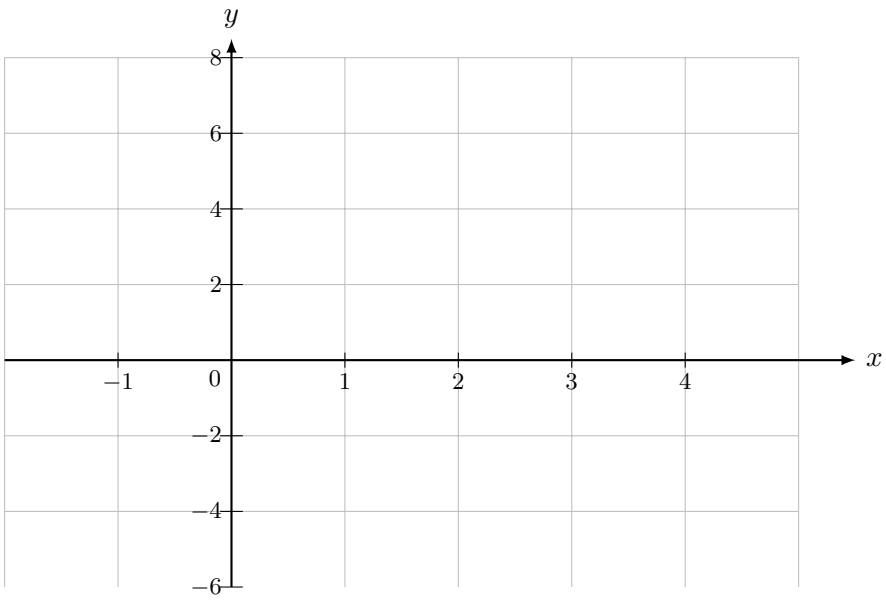
..... **[1]**

(c) Find the coordinates of the points of intersection of the two graphs.

Point 1 = ... , Point 2 = ... **[3]**

11 Let $f(x) = x^3 - 4x^2 + x + 2$ for the domain $-1 \leq x \leq 4$.

(a) On the axes, sketch the graph of $y = f(x)$.



[2]

(b) Write down the x-coordinates of the points where $f(x) = 0$.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[3]**

(c) Write down the coordinates of the local minimum.

(..... , ...) **[2]**

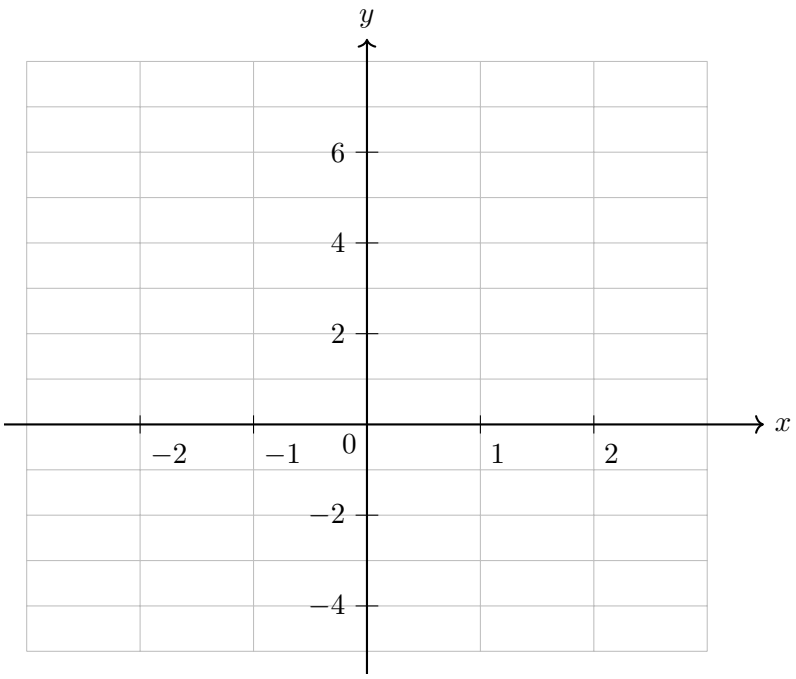
(d) On the same axes, sketch the graph of $g(x) = -2x + 1$. **[1]**

(e) Find the x-coordinates of the points of intersection of the graphs of $f(x)$ and $g(x)$.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[3]**

12 Let $f(x) = x^3 - 3x + 2$.

(a) On the axes, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.



[3]

(b) Find the coordinates of the local maximum.

(..... , ...) **[2]**

(c) Find the zeros of $f(x)$.

$x =$ ..... , $x =$ ..... **[2]**

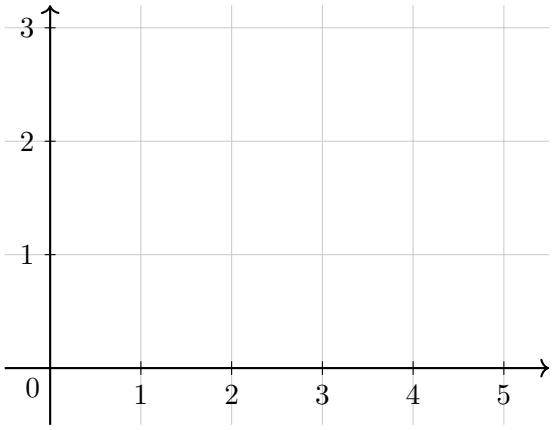
(d) On the same diagram, sketch the graph of $y = 2x + 1$. **[2]**

(e) Find the x-coordinates of the points of intersection of $y = f(x)$ and $y = 2x + 1$.

..... **[3]**

13 Let $f(x) = 3x \cdot 2^{-x}$ and $g(x) = 0.5x + 0.2$ for $0 \leq x \leq 5$.

(a) On the axes provided, sketch the graph of $y = f(x)$ and the graph of $y = g(x)$ for $0 \leq x \leq 5$.



[3]

(b) Find the coordinates of the local maximum of $y = f(x)$.
Give your answer correct to 2 decimal places.

(..... , ...) **[2]**

(c) Solve the inequality $f(x) > g(x)$ for $0 \leq x \leq 5$.
Give your answer correct to 2 decimal places.

..... **[2]**

14 Consider the function $h(x) = x^3 - 3x^2 - 2x + 5$ for $-2 \leq x \leq 4$.

(a) Use your graphic display calculator to find the zeros of $h(x)$.
Give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

(b) Find the coordinates of the local maximum point and the local minimum point of the graph of $y = h(x)$.
Give your answers correct to 2 decimal places.

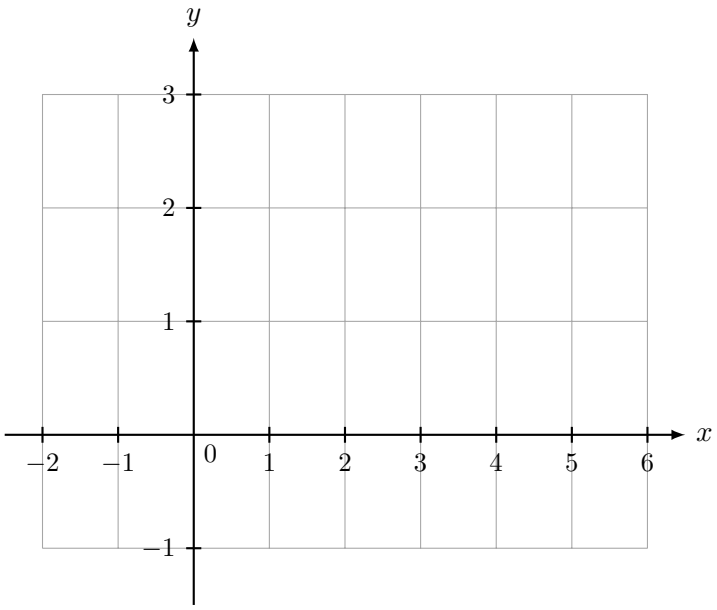
Local maximum = ... , Local minimum = ... **[2]**

(c) The equation $h(x) = k$ has three distinct real solutions.
Write down the range of possible values of k, giving your boundary values correct to 2 decimal places.

..... **[2]**

15 The function $f(x)$ is defined as $f(x) = \dfrac{x^2}{2^x}$ for the domain $-1 \leq x \leq 5$.

(a) On the grid, sketch the graph of $y = f(x)$.



[3]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(c) Find the x-coordinates of the points where $f(x) = 0.5$.

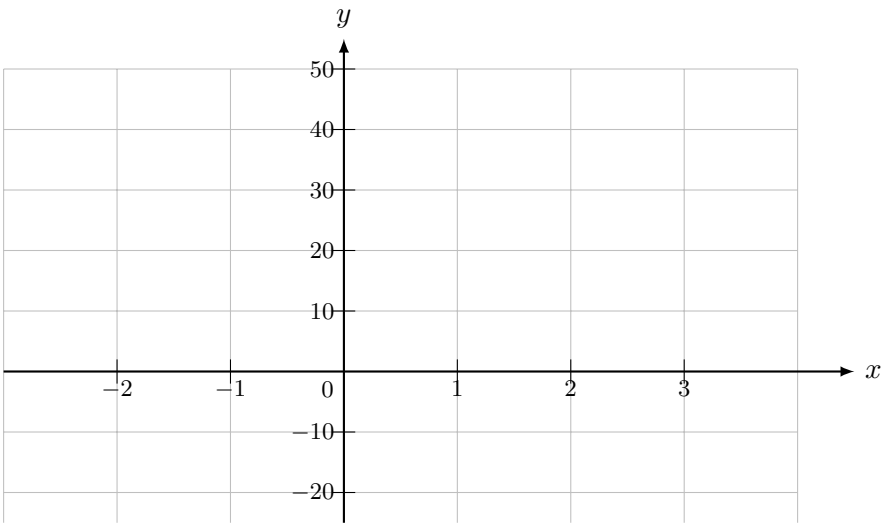
$x =$ ..... **[2]**

(d) Write down the range of values of k for which the equation $f(x) = k$ has exactly 3 different solutions.

..... **[2]**

16 The function $f(x)$ is defined as $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x$ for the domain $-2 \leq x \leq 3$.

(a) On the axes provided, sketch the graph of $y = f(x)$.



[3]

(b) Write down the coordinates of the two local minima.

.....

..... **[2]**

(c) Write down the coordinates of the local maximum.

(..... , ...) **[1]**

(d) The equation $f(x) = k$ has exactly four distinct solutions.
Write down the range of values for k.

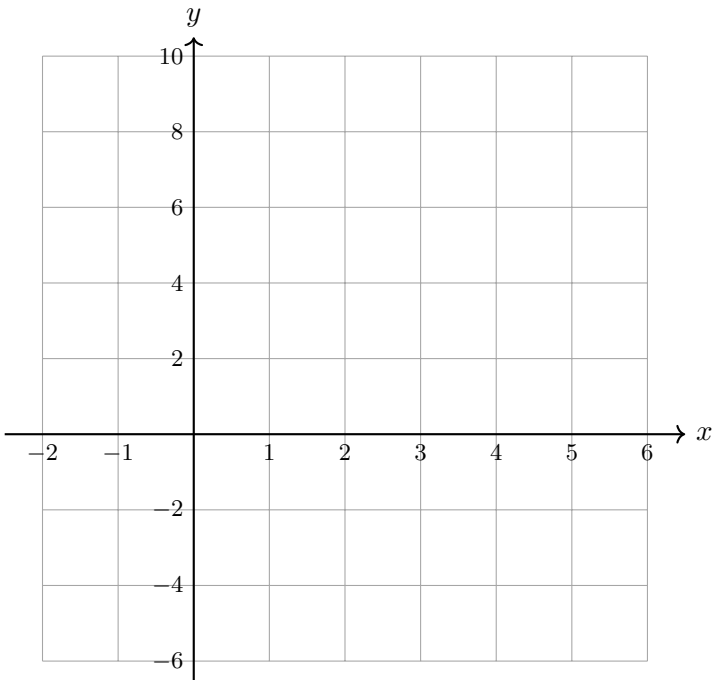
..... **[2]**

(e) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$.
Write down the coordinates of the local maximum of the translated graph.

(..... , ...) **[1]**

17 Let $f(x) = \dfrac{x^2 - 2x + 4}{x - 2}$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 6, x \neq 2$.



[3]

(b) Write down the equation of the vertical asymptote.

..... **[1]**

(c) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(d) Find the coordinates of the local minimum point.

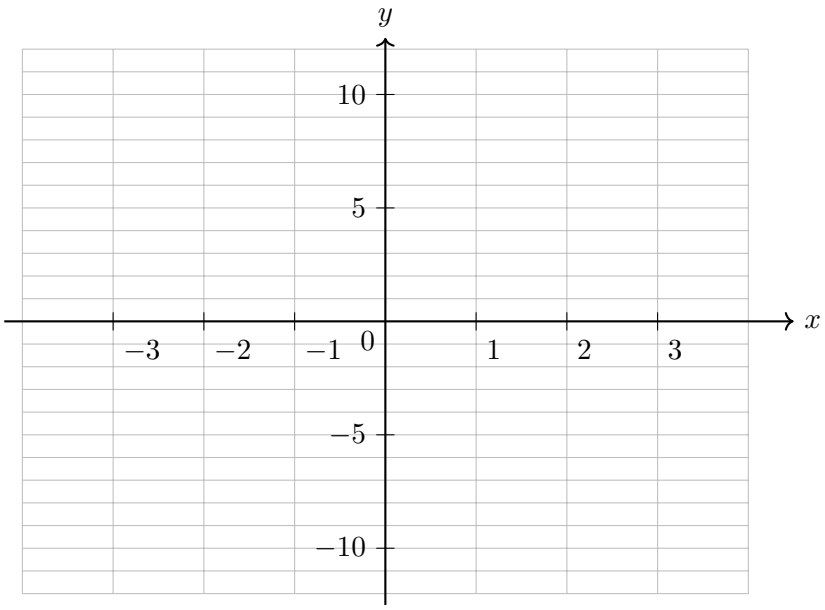
(..... , ...) **[2]**

(e) The equation $f(x) = k$ has no real solutions.
Write down the range of values of k.

..... **[2]**

18 Let $f(x) = \dfrac{x^3 + 2}{x}$.

(a) On the axes, sketch the graph of $y = f(x)$ for $-3 \leq x \leq 3$.



[3]

(b) Find the coordinates of the local minimum.

(..... , ...) **[2]**

(c) Find the zero of $f(x)$.

$x =$ ..... **[1]**

(d) Write down the equation of the vertical asymptote.

..... **[1]**

(e) Solve the equation $f(x) = 4$.

..... **[3]**